Combining terms of like powers of h, we have

$$P = p + hF + \frac{h^2}{2}pF' + \frac{h^3}{4}\left[F'F + p^2F''\right] + \mathcal{O}(h^4).$$

We carry these terms to compare against the expansion of the exact solution. Since $\dot{q} = p$, we have $\ddot{q} = \dot{p} = F(q)$, and the third derivative is $q^{(3)} = F'(q)p$. On the other hand $\dot{p} = F(q)$ implies that $\ddot{p} = F'(q)\dot{q} = pF'$, and thus

$$p^{(3)} = p^2F'' + F'F.$$

The Taylor expansion of the solution is (taking $q(t) = q, p(t) = p$):

$$q(t+h) = q + hp + \frac{h^2}{2}F + \frac{h^3}{6}F'p + \mathcal{O}(h^4),$$
$$p(t+h) = p + hF + \frac{h^2}{2}pF' + \frac{h^3}{6}\left[p^2F'' + F'F\right] + \mathcal{O}(h^4).$$

We now examine the series expansions for the exact and Verlet solutions and find that these differ in the third (and higher) order terms.

$$Q - q(t+h) = \frac{h^3}{6}F'p + \mathcal{O}(h^4),$$

and

$$P - p(t+h) = \frac{h^3}{12}\left[p^2F'' + F'F\right] + \mathcal{O}(h^4).$$

These relations can be summarized as telling us that

$$\|\mathscr{G}_h(\boldsymbol{z}) - \mathscr{F}_h(\boldsymbol{z})\| = \kappa(\boldsymbol{z})h^3 + \mathcal{O}(h^4),$$

where $\kappa(\boldsymbol{z}) = \kappa(q,p)$ is a function of the position and momentum.

We may then define

$$\bar{K} = \max_{t\in[0,\tau]} \kappa(\boldsymbol{z}(t))$$

bounding the local error by (with neglect of the fourth order terms) $\bar{K}h^3$. Thus the Verlet method is consistent of order two.

To complete the convergence proof for Verlet's method, we would still need to verify the second assumption. This requires the assumption that the force field $\boldsymbol{F}$ satisfy a Lipschitz condition:

$$\|\boldsymbol{F}(\boldsymbol{u}) - \boldsymbol{F}(\boldsymbol{w})\| \leq \hat{L}\|\boldsymbol{u} - \boldsymbol{w}\| \tag{2.15}$$